

Verification status

Repository source: VERIFICATION_STATUS.md

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Version: v1.1.0 **Release date:** September 26, 2026 **Status date:** September 26, 2026 **Audited pre-release commit:** 892d4bc6fad07950e78da66e45b95063a6415af6 **Audited hosted workflow:** <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>

Proposed theorem

For every convex body $K \subset \mathbb{R}^3$ of constant width $d > 0$, the package represents the proposed universal inequality

$$\text{Vol}(K) \geq \frac{26221074253\pi}{200000000000} d^3 = 0.41187967121228637754323153860 \dots d^3.$$

Current designation

Proposed exact universal certificate - standalone release v1.1.0.

The finite exact verifier, independent binary64 Python audit, independent base-R audit, regression tests, adversarial mutation tests, manuscript reconciliation, documentation checks, hosted three-job workflow, and named- author fresh-clone replay pass in the recorded environments. The result has not been externally reproduced, peer reviewed, or accepted for journal publication.

Publication metadata

- sole named author: Bruce J. Swihart;
- institutional affiliation: none asserted;
- canonical repository: <https://github.com/swihart/minvol-degree3-contact-bound>;
- versioned release: <https://github.com/swihart/minvol-degree3-contact-bound/releases/tag/v1.1.0>;
- version: v1.1.0;
- release date: September 26, 2026;
- prose license: CC BY 4.0;
- software license: MIT;
- archival DOI: none assigned or implied.

Verification matrix

ID	Verification item	Evidence	Status
E1	Exact lower-row, fan, determinant, coordinate-width, cap, and handoff inequalities	<code>certificate/certify_universal_bound.py</code> ; certificate JSON; exact transcript	Passed
E2	Byte-identical reconstruction of final machine-readable certificate	Production exact verifier	Passed
N1	Independent binary64 reconstruction	<code>certificate/audit_universal_bound.py</code>	Passed locally and hosted
R1	Independent base-R reconstruction	<code>certificate/audit_universal_bound.R</code> ; named-author and hosted transcripts	Passed locally and hosted
T1	Reconstruction and regression suite	Ten Python tests	Passed locally and hosted
M1	Deliberate corruption detection	Seven adversarial mutation tests	Passed locally and hosted
P1	Paper coefficient, split, lower rows, and main constants match certificate	<code>paper/reconcile_certificate.py</code>	Passed locally and hosted
D1	Exact-constants appendix matches certificate	<code>proof/generate_exact_constants.py</code>	Passed locally and hosted
D2	Required release documents and relative links are present	<code>ci/verify_release_docs.py</code>	Passed locally and hosted
D2a	Author, repository, version, date, citation, and split-license metadata are internally consistent	<code>ci/preflight_metadata.sh</code> ; <code>ci/preflight_release.sh</code> ; <code>CITATION.cff</code> ; <code>LICENSE</code>	Passed in release preparation
D3	Every repository Markdown source has a checksummed PDF counterpart	<code>ci/verify_doc_pairs.py</code>	22 of 22 passed locally and hosted
P2	Paper and documentation PDFs are readable, have embedded fonts, and expose extractable text	Hosted Poppler preflight on committed and rebuilt PDFs	Passed on run 35468019663 before the final metadata transition; repeated by the tag workflow before publication
C1	Fresh standalone replay with no private path dependency	Named-author remote clone of commit 892d4bc6...	Passed; final status clean
C2	Green hosted Python, R, and document jobs on the audited candidate	Three jobs on commit 892d4bc6...	Passed on run 35468019663
C3	Complete public-history bundle and transfer hashes verified	Bundle and three transferred evidence artifacts	Passed
C4	Hosted workflow and fresh-clone evidence are internally indexed	<code>certificate/evidence/rc1_remote_audit.json</code> ; <code>ci/verify_remote_evidence.py</code>	Passed

ID	Verification item	Evidence	Status
C5	Green hosted jobs and fresh replay on the immutable final tag	Final v1.1.0 tag and release procedure	Publication gate
A1	Deterministic release assets and SHA-256 manifest verify	<code>release/build_release_assets.py</code> ; <code>release/verify_release_assets.py</code>	Preview-tested; tagged build required for publication
X1	Independent mathematical reproduction by an outside reviewer	None yet	Pending
J1	Peer review or journal acceptance	None yet	Pending

Recorded pre-release evidence

The audited pre-release commit is

```
892d4bc6fad07950e78da66e45b95063a6415af6
Add release-candidate citation and licensing metadata
```

The hosted workflow at <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663> checked out that commit in all three jobs. The downloaded log archive contains the expected exact, binary64, ten-test, seven-mutation, release-document, metadata, paper, base-R, Markdown/PDF-pair, and Poppler-preflight pass markers.

A separate fresh clone reproduced the complete Python and base-R path under Python 3.14.7, NumPy 2.3.5, and R 4.6.1. It rebuilt the ten-page paper and all 22 Markdown/PDF pairs and ended with an empty `git status --short`. The raw transfer artifacts are identified by SHA-256 in [CLEAN_CLONE_CHECK.md](#) and the machine-readable evidence index at [certificate/evidence/rc1_remote_audit.json](#). That record remains intentionally labeled `rc1`: it documents the audited pre-release state rather than rewriting history after the final version transition.

What “exact” means here

The theorem-bearing finite comparisons use Python integers and `fractions.Fraction`. The verifier reads immutable integer arrays, checks the semantic inequalities, reconstructs derived artifacts byte for byte, and compares exact rational coefficients. No binary floating-point inequality is used as theorem authority.

“Exact” does not mean that the whole research claim has been independently validated. The proof still depends on the correctness and completeness of the mathematical reduction encoded in the manuscript and verifier, the trusted software/hardware stack, and the absence of a shared conceptual error.

See [certificate/TRUST_BOUNDARY.md](#).

What is established inside the package

- gap-free circumradius coverage from $1/2$ to $\sqrt{6}/4$;
- lower-branch weakest row `03B02B`; the universal bottleneck is the near-Jung branch;
- exact refinement of the formerly weak lower-row interval;
- final fan reconstruction with 98,306 rays;
- positive exact fan margin;
- determinant transfer and dominant-edge coordinate-width bounds;
- forced-core and three cap-pair lower bounds;
- exact six-cap disjointness;
- a near-Jung lower bound that sets the universal bottleneck;
- exact universal coefficient assembly;
- paper and appendix reconciliation with the certificate;
- rejection of seven representative proof-object corruptions; and

- successful hosted and fresh-clone replay of the audited pre-release state.

What is not established

The package does not establish:

- sharpness, equality, or rigidity;
- the identity of a minimizing body;
- the Meissner conjecture;
- external independent mathematical reproduction;
- peer review, journal acceptance, or an archival DOI;
- an exhaustive priority or novelty determination.

The Meissner volume is used only as an explicit-body comparison value.

Publication boundary

The repository is prepared for release `v1.1.0`. Publication of the GitHub Release is conditioned on all of the following operational checks:

1. the final release commit is clean and receives three green hosted jobs;
2. the annotated `v1.1.0` tag points to that exact commit;
3. the tag-triggered Python, R, and document jobs are green;
4. a fresh tagged checkout completes the full replay and ends clean;
5. tagged release assets are built, hashed, downloaded, and reverified; and
6. the named author explicitly approves the visibility change, GitHub Release, and announcement text.

External review remains a scientific goal after release and must not be implied by a green workflow, named-author fresh-clone replay, or exact certificate.